

Figure 1

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1 Packages, data and functions

1.1 Packages

```
library(tidyverse)
library(patchwork)
library(ggstance)
library(here)
library(lme4)
library(lmerTest)
library(kableExtra)
```

1.2 Data

```
load(here("Data/3_ForecastsResults/3_MergedResults/complete.forecasts.RData"))
```

1.3 Functions

```
conf_interval <- function(model, nd, response, renameResponse, cmult=1.96){
  nd[,response] <- predict(model, nd, re.form=NA)
```

```

mm <- model.matrix(terms(model),nd)
pvar1 <- diag(mm %*% tcrossprod(vcov(model),mm))
nd <- nd %>%
  dplyr::mutate(lower=!!as.name(response)-cmult*sqrt(pvar1),
               upper=!!as.name(response)+cmult*sqrt(pvar1),
               Response=renameResponse) %>%
  dplyr::rename(fit=!!as.name(response))
return(nd)
}

```

2 Figure 1 & Table 1

2.1 Models

```

## taxa
formula <-
  as.formula(
    RMSE ~ median_richness_center * light_treatment +
      (1 | Target) + (1 | bottle) + (1 | incubator)
  )

m.simplex <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(
      forecast.method == "Simplex EDM",
      Target != "Total_biomass",
      Target != "Mean_oxygen"
    )
)

## biomass
formula <-
  as.formula(RMSE ~ median_richness_center + light_treatment +
    (1 | composition) + (1 | incubator))

```

```

m.simplex.bio <- lmer(
  formula,
  complete.forecasts %>%
    dplyr::filter(forecast.method == "Simplex EDM",
                  Target == "Total_biomass")
)

```

2.2 Figure

```

dd.m.simplex <- complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM",
                Target != "Total_biomass",
                Target != "Mean_oxygen")

newdat.m.simplex <-
  expand.grid(
    median_richness_center = seq(
      min(dd.m.simplex$median_richness_center),
      max(dd.m.simplex$median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(dd.m.simplex$light_treatment)
  )

newdat.m.simplex <-
  conf_interval(m.simplex, newdat.m.simplex, "RMSE", "Forecast error (RMSE)")

nbot <- complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM") %>%
  group_by(median_richness_center, light_treatment, bottle) %>%
  summarize(nspecies = n()) %>%
  group_by(median_richness_center, light_treatment) %>%
  summarize(nbottles = paste0("n=", n()))

figure1a <- complete.forecasts %>%
  dplyr::filter(Target != "Total_biomass",

```

```

        Target != "Mean_oxygen",
        forecast.method == "Simplex EDM") %>%
ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
geom_ribbon(
  data = newdat.m.simplex,
  aes(
    ymin = lower,
    ymax = upper,
    y = fit,
    group = light_treatment
  ),
  col = NA,
  fill = "grey",
  alpha = 0.3
) +
geom_line(data = newdat.m.simplex,
          size = 1.3,
          aes(y = fit, group = light_treatment)) +
geom_point(size = 2.5, alpha = 0.6) +
labs(x = "Centered median realized species richness",
     y = "Taxa abundance forecast error (RMSE)",
     col = "Light conditions") +
scale_color_manual(values = c("#C4961A", "#293352"),
                   aesthetics = c("color", "fill")) +
theme_bw() +
geom_text(
  data = nbot,
  aes(x = median_richness_center, y = 3, label = nbottles),
  position = position_dodgev(height = 0.2),
  fontface = "bold",
  show.legend = F,
  # position = position_dodge(width = 0.5),
  size = 3.5
) +
ylim(0, 3.05)

```

```

#-----
#### Fig1b: Biomass forecast simplex
#-----

dd.m.simplex <- complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM",
                Target == "Total_biomass")

newdat.m.simplex.bio <-
  expand.grid(
    median_richness_center = seq(
      min(dd.m.simplex$median_richness_center),
      max(dd.m.simplex$median_richness_center),
      length.out = 100
    ),
    light_treatment = unique(dd.m.simplex$light_treatment)
  )

newdat.m.simplex.bio <-
  conf_interval(m.simplex.bio,
                newdat.m.simplex.bio,
                "RMSE",
                "Forecast error (RMSE)")

figure1b <- complete.forecasts %>%
  dplyr::filter(Target == "Total_biomass", forecast.method == "Simplex EDM") %>%
  ggplot(aes(median_richness_center, RMSE, col = light_treatment)) +
  geom_ribbon(
    data = newdat.m.simplex.bio,
    aes(
      ymin = lower,
      ymax = upper,
      y = fit,
      group = light_treatment
    ),
    col = NA,

```

```

    fill = "grey",
    alpha = 0.3
  ) +
  geom_line(data = newdat.m.simplex.bio,
            size = 1.3,
            aes(y = fit, group = light_treatment)) +
  geom_point(size = 2.5, alpha = 0.6) +
  labs(x = "Centered median realized species richness",
       y = "Community biomass forecast error (RMSE)",
       col = "Light conditions") +
  scale_color_manual(values = c("#C4961A", "#293352"),
                    aesthetics = c("color", "fill")) +
  theme_bw() +
  ylim(0, 3.05)

```

```

figure1a + figure1b +
  plot_layout(guides = "collect") + plot_annotation(tag_levels = "a")

```

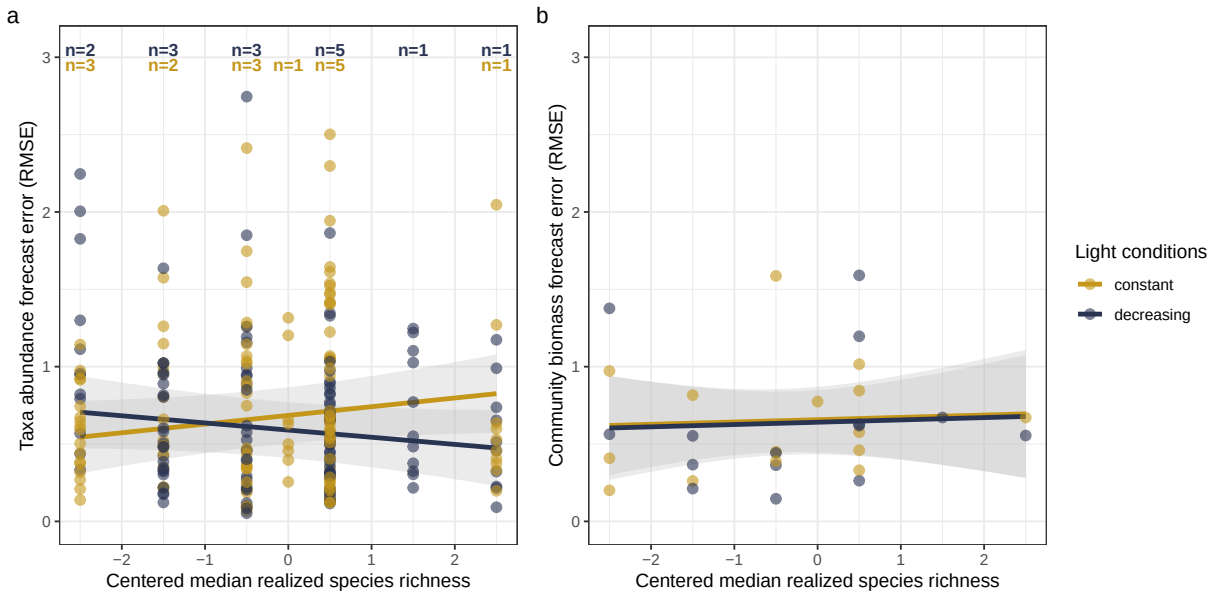


Figure 1: Effects of species richness and light on forecast error. The two treatments had interactive effects on forecast error of taxa abundances (a) but no effect on the forecast error of community biomass (b). Forecasts were computed with Simplex Empirical Dynamic Modeling (EDM), and forecast error was calculated as root mean square error (RMSE). RMSE is standardized (unitless). Observed richness deviated from planned

richness because not all species persisted throughout the experiment. We therefore used realized richness (i.e. the median of the observed richness values) as explanatory variable. Median richness was centered; a centered median richness of zero corresponds to a median richness of 5.5 (minimum median richness: 3; maximum median richness: 8). Lines display the fit of the linear mixed model; shaded areas denote the 95% confidence intervals. In (a), we forecasted 253 taxa time series (across all bottles); each point represents a forecasted taxon in a bottle. Taxon, bottle, and incubator were included as random intercepts in the model. Inserts at the top indicate the number of bottles for the respective richness level (yellow: constant light, blue: decreasing light). In (b), we forecasted community biomass in 30 bottles. Composition and incubator were included as random intercepts in the model.

2.3 Table

```
#### -----
#### Table for simplex species forecasts regression (Main text)
#### -----

conf.int.1 <- confint(m.simplex)
conf.int.1 <- conf.int.1[5:8, ]
conf.int.1 <- round(conf.int.1, 3)
conf.int.1 <- apply(conf.int.1, 1, paste, collapse = " - ")

spec.ab.tab <-
  data.frame(
    Response = "Forecast error\nof target\nabundances",
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)",
      "Richness:Light"
    ),
    Estimate = fixef(m.simplex),
    CI = conf.int.1,
    SS = c(NA, anova(m.simplex)[, 1]),
    DF = c(NA, paste(
      anova(m.simplex)[, 3], round(anova(m.simplex)[, 4], 1), sep = ", "
    )),
```

```

    F.value = c(NA, anova(m.simplex)[, 5]),
    p.value = c(NA, anova(m.simplex)[, 6])
  )

#### -----
#### Table for simplex community biomass regression (Main text)
#### -----

conf.int.2 <- confint(m.simplex.bio)
conf.int.2 <- conf.int.2[4:6, ]
conf.int.2 <- round(conf.int.2, 3)
conf.int.2 <- apply(conf.int.2, 1, paste, collapse = " - ")

bio.rmse.tab <-
  data.frame(
    Response = "Forecast error\nof community\nbiomass",
    Covariate = c(
      "Intercept",
      "Richness",
      "Light (decreasing)"
    ),
    Estimate = fixef(m.simplex.bio),
    CI = conf.int.2,
    SS = c(NA, anova(m.simplex.bio)[, 1]),
    DF = c(NA, paste(
      anova(m.simplex.bio)[, 3], round(anova(m.simplex.bio)[, 4], 1), sep = ", "
    )),
    F.value = c(NA, anova(m.simplex.bio)[, 5]),
    p.value = c(NA, anova(m.simplex.bio)[, 6])
  )

main.tab <- rbind(spec.ab.tab, bio.rmse.tab)

rownames(main.tab) <- NULL

main.tab$p.value <- ifelse(main.tab$p.value < 0.0001, "<0.0001",

```

```

        sprintf("%.4f", round(main.tab$p.value, 4)))

main.tab <- main.tab %>%
  dplyr::mutate(
    Estimate = round(Estimate, 3),
    SS = round(SS, 3),
    F.value = round(F.value, 3)
  )

options(knitr.kable.NA = '')
main.tab %>%
  mutate_all(linebreak, align = "l") %>%
  kable(
    "latex",
    booktabs = T,
    digits = 3,
    col.names = c(
      "Response",
      "Covariate",
      "Estimate",
      "95\\% CI",
      "SS",
      "DF",
      "\\textit{F}-value",
      "\\textit{p}-value"
    ),
    caption = "Results of two linear mixed models investigating the effects of
species richness and light on the forecast error of taxa abundances and
community biomass, respectively. We forecasted 253 taxa time series (across
all bottles) and community biomass in 30 bottles. In both models, median
centered richness and light treatment were fixed effects. The first model
(response: forecast error of taxa abundances) included taxon, bottle, and
incubator as random intercepts. The second model (response: forecast error
of community biomass) included composition and incubator as random
intercepts. Forecast error is based on forecasts computed with Simplex
Empirical Dynamic Modeling (EDM). In the second model, the interaction

```

Table 1: Results of two linear mixed models investigating the effects of species richness and light on the forecast error of taxa abundances and community biomass, respectively. We forecasted 253 taxa time series (across all bottles) and community biomass in 30 bottles. In both models, median centered richness and light treatment were fixed effects. The first model (response: forecast error of taxa abundances) included taxon, bottle, and incubator as random intercepts. The second model (response: forecast error of community biomass) included composition and incubator as random intercepts. Forecast error is based on forecasts computed with Simplex Empirical Dynamic Modeling (EDM). In the second model, the interaction was not significant (p -value > 0.1) and was therefore removed from the model. The last four columns report the ANOVA results. P -values < 0.05 in bold. See Supplement Table S6 for further details.

Response	Covariate	Estimate	95% CI	SS	DF	F -value	p -value
Forecast error of target abundances	Intercept	0.685	0.503 - 0.865				
	Richness	0.056	-0.008 - 0.121	0.008	1, 21.1	0.046	0.8328
	Light (decreasing)	-0.095	-0.265 - 0.081	0.225	1, 6.1	1.230	0.3092
	Richness:Light	-0.102	-0.193 - -0.014	0.922	1, 19.7	5.046	0.0363
Forecast error of community biomass	Intercept	0.657	0.447 - 0.866				
	Richness	0.015	-0.107 - 0.134	0.004	1, 19.8	0.058	0.8118
	Light (decreasing)	-0.017	-0.23 - 0.186	0.002	1, 14.1	0.028	0.8698

```

was not significant (p-value > 0.1) and was therefore removed from the
model. The last four columns report the ANOVA results. P-values < 0.05
in bold. See Supplement Table S6 for further details. ",
align = c("llcccr"),
escape = F,
label = "MainTab"
) %>%
collapse_rows(
  columns = c(1),
  latex_hline = 'custom',
  custom_latex_hline = c(1),
  valign = "middle"
) %>%
kable_styling(font_size = 7)

```

3 Versions of R and of the R-packages used

R version: R version 4.4.1 (2024-06-14 ucrt).

Base packages: stats, graphics, grDevices, utils, datasets, methods, base.

Other packages: kableExtra 1.4.0, lmerTest 3.1-3, lme4 1.1-36, Matrix 1.7-0, here 1.0.1, ggstance 0.3.7, patchwork 1.3.0, lubridate 1.9.4, forcats 1.0.0, stringr 1.5.1, dplyr 1.1.4, purrr 1.0.4, readr 2.1.5, tidyr 1.3.1, tibble 3.2.1, ggplot2 3.5.1, tidyverse 2.0.0, knitr 1.49.

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